Northumbria University NEWCASTLE

Module: Enterprise Networks and Security

# Section 1. Network decisions and justification

| Site | Router | Interface | Connects To | IP Address |
|---|---|---|---|---|
| **Manchester HO** | EXEC | Fa0/0 | MAN_HO | 10.1.10.2/30 |
| **Manchester HO** | EXEC | Fa0/1 | EXEC LAN | 10.1.1.1/26 |
| **Manchester HO** | EXEC | Fa2/0 | HR (direct) | 10.1.12.1/30 |
| **Manchester HO** | HR | Fa0/0 | MAN_HO | 10.1.11.2/30 |
| **Manchester HO** | HR | Fa0/1 | HR LAN | 10.1.20.1/24 |
| **Manchester HO** | HR | Fa2/0 | EXEC (direct) | 10.1.12.2/30 |
| **Manchester HO** | MAN_HO | Fa0/0 | HR | 10.1.11.1/30 |
| **Manchester HO** | MAN_HO | Fa0/1 | EXEC | 10.1.10.1/30 |
| **Manchester HO** | MAN_HO | Fa2/0 | P Router | 172.20.20.2/30 |
| **MPLS Core** | P | Fa0/0 | MAN_HO | 172.20.20.1/30 |
| **MPLS Core** | P | Fa0/1 | LDS | 172.20.20.5/30 |
| **MPLS Core** | P | Fa2/0 | NCL | 172.20.20.9/30 |
| **Leeds** | LDS | Fa0/1 | PRODUCTION1 | 10.2.10.1/30 |
| **Leeds** | LDS | Fa2/0 | P Router | 172.20.20.6/30 |
| **Leeds** | PRODUCTION1 | Fa0/1 | LAN / Switch | 10.2.1.1/25 |
| **Leeds** | PRODUCTION1 | Fa0/0 | NCL | 10.2.10.2/30 |
| **Newcastle** | NCL | Fa0/0 | LOGISTICS | 10.3.12.1/30 |
| **Newcastle** | NCL | Fa0/1 | R&D | 10.3.10.1/30 |
| **Newcastle** | NCL | Fa1/0 | P Router | 172.20.20.10/30 |
| **Newcastle** | NCL | Fa2/0 | PRODUCTION2 | 10.3.11.1/30 |
| **Newcastle** | R&D | Fa0/0 | NCL | 10.3.10.2/30 |
| **Newcastle** | R&D | Fa0/1 | LAN A | 10.3.1.1/26 |
| **Newcastle** | R&D | Fa2/0 | LAN B | 10.3.2.1/28 |
| **Newcastle** | PRODUCTION2 | Fa0/0 | NCL | 10.3.11.2/30 |
| **Newcastle** | PRODUCTION2 | Fa0/1 | LAN / Switch | 10.3.3.1/23 |

| | | | | |
|---|---|---|---|---|
| **Newcastle** | LOGISTICS | Fa0/0 | LAN / Switch | 10.3.5.1/25 |
| **Newcastle** | LOGISTICS | Fa0/1 | NCL | 10.3.12.2/30 |

## Requirement 1:

The EXEC and HR departments at the Head Office (MAN_HO) are responsible in exchanging sensitive information in the organization through the Ethernet. As this connection will contain confidential data, every signal should be guarded against being hacked, tampered with, and hacked. To this end, an IPsec VPN was installed between EXEC and HR routers to encrypt all direct connection traffic.

The reason why IPsec was selected is that it offers network-layer security in the form of confidentiality, integrity, authentication, and replay-protection without being exposed to the end-user applications. IPsec is optimal in intra-departmental links when a number of services can be operated without application level security restrictions. All the EXEC-HR traffic is encrypted by IPsec, and sensitive information remains confidential even in case of interception (Mungse, 2025). The use of IPsec was also preferable as compared to the use of the SSL/TLS as IPsec always secures the IP traffic and it does not require any protocol-independent security parameters (Frankel et al., 2005).

IKE secures negotiating router secure keys. IKE also makes the cryptographic key management easier and allows creating Security Associations with security that do not distribute the keys physically, thereby minimizing the human error (Mungse, 2025). AES algorithm is employed in encryption since it is well-known, effective, and safe. It is also appropriate to use in the enterprise as AES can utilize numerous key sizes and works more effectively than DES and 3DES (Abdullah, 2017).

SHA using HMAC ensured integrity and authenticity of data. SHA has been selected because it has excellent resistance to cryptographic attacks and is more secure than weaker hash functions (Oj, 2015). The old-fashioned hash hash algorithm (SHA-1) can be used in making HMAC-based integrity protection on small internal environment with minimal exposure to attackers and possessing great security as compared to its discarded rivals (Oj, 2015).

In phase 1 of IKE, the peers were authenticated through PSK. PSK based authentication suits small networks that are internally operated and have a small number of VPN peers whereas PK cryptography is global. PSK streamlines operations and achieves EXEC-HR authentication (Perez, 2025).

The selected IPsec-based solution is a good tradeoff between security and manageability. the implementation will work to encrypt all EXEC-HR Ethernet traffic and apply common cryptographic technologies to protect valuable communications within the departments to meet the security requirements of the organization.

## Requirement 2:

The reason why MPLS layer 3 VPN was chosen to link MAN_HO HR, LDS PRODUCTION1 and NCL LOGISTICS is that it is the most scalable and has a high level of management efficiency than any other solution. Compared to a standard IPsec mesh deployment, which would have to make n(n-1)/2 = 3 tunnel setups, MPLS VPN provides the same degree of connectivity with centralized PE router setup (Sofi and Gurm, 2015). The label forwarding system gets rid of recursive IP routing within every hop, and allows wire-speed forwarding of packets across the MPLS backbone at much lower latency than with traditional IP routing (Mungse, 2025). To ensure that the logical traffic separation of the other departments is achieved, a special VRF called CUSTOMER_A on all three PE routers (MAN_HO, LDS, NCL) was created. VRF technology allows supporting multiple, isolated routing instances, each VRF having its own Routing Information Base (RIB), Forwarding Information Base (FIB), and Label Forwarding Information Base (LFIB) (Sofi and Gurm, 2015). This isolation secures that HR, PRODUCTION1 and LOGISTICS traffic is no longer impacted by any UNINTENT or malicious inter-VPN traffic.

All PE routers had different Route Distinguishers: MAN_HO (65000:100), LDS (65000:200) and NCL (65000:300). RDs adjust normal IPv4 paths to VPNv4 paths with an added 64-bit identifier to avoid address conflicts; they do this by ensuring that identical private IP groups are not shared among separate customer VPNs (Talaulikar et al., 2025). The format of RD is ASN:index, in which 65000 is the number of the private Autonomous System of the enterprise MPLS implementation.

The three PE routers had the same Route Targets (export/import 65000:100). This RT architecture generates automatic full mesh connectivity: routes exported with RT 65000:100 by any PE router are selectively imported by all other PE routers with matching import policy, meaning that direct any-to-any communication can be made between the three sites without the hub-and-spoke restrictions of hub-and-spoke routing (Mungse, 2025).

PE routers attached with Mp-BGP with VPNv4 address family were used in order to distribute customer routes along with their respective RD and RT attributes across the MPLS backbone. The MP-BGP builds on the BGP by transmitting VPN routing data and guaranteeing a high level of VRF isolation (Dharun et al., 2024). PE routers use their loopback addresses (2.2.2.2, 3.3.3.3, 4.4.4.4) in establishing iBGP sessions instead of physical interface IPs which makes sessions more stable in the event of physical link failures or maintenance procedures.

The ability to implement full mesh was confirmed by extensive bidirectional reachability testing. Ping tests to the remote sites by VRF-aware ping tests by all three PE routers had 100 percent success rates, indicating that MPLS label-switched paths accurately route VPN traffic in all directions across the provider backbone. The any-to-any connectivity, which has been tested, confirms that the full mesh topology allows direct communication among the HR, PRODUCTION1 and LOGISTICS departments as indicated in Requirement 2.

## Requirement 3:

This is carried out to provide a hybrid security architecture whereby multiplexed layer 3 VPN is merged with selective IPsec encryption between the EXEC and the R&D location. The design uses VRF CUSTOMER_B (RD 65000:400 at MAN_HO, RD 65000:500 at NCL) with equivalent route targets (RT 65000:200) to create a special VPN tunnel that is independent of any other network traffic. The MPLS VPN mesh networking suggests that any two network nodes may be connected directly, and the confidentiality, integrity, and availability of data transmission are guaranteed (Chen, 2025). MP-BGP will redistribute connected interfaces and redistribute the static routes which permits the PE routers to exchange routes automatically.

The underlying transport technology that is chosen is MPLS Layer 3 VPN because it can offer scalable and private connectivity in a service provider setting without cryptographic overhead. The previous studies emphasize that MPLS VPNs provide efficient forwarding and predictable performance by executing the packet classification at the network edge and keeping traffic isolation throughout the core (Chen, 2025). This renders MPLS especially applicable to enterprise settings where several departments must have regulated inter-site communication.

Although MPLS VPNs provide logical isolation, they do not encrypt payload data. This is why; extra IPsec protection was made selectively to the traffic between EXEC and R&D LAN A where there is more sensitive information. The Customer Edge router implemented IPsec instead of the Provider Edge because site-to-site IPsec is intended to create authenticated and encrypted connections between trusted enterprise endpoints and does not rely on the service provider to handle encryption keys (Bilhaj et al., 2026). This will maintain the transparency of providers, maintain confidentiality and integrity on sensitive information.

The interesting traffic was defined based on selective encryption by using extended access control lists to only encrypt communication between EXEC LAN (10.1.1.0/26) and R&D LAN A (10.3.1.0/26). The IPsec explicitly excluded R&D LAN B (10.3.2.0/ 28), because non-sensitive traffic must not be subjected to an unreasonable security cost. Research has demonstrated that too much cryptographic tunnelling may add more delay and processing overhead to the edge devices, and selective deployment is an effective design decision (Bilhaj et al., 2026).

Overall, this design satisfies Requirement 3 by combining direct MPLS VPN connectivity with selective IPsec protection, achieving an appropriate balance between security and performance

## Requirement 4:

This requirement involves determining the optimal placement for a company-internal SQL server that provides query services exclusively to users within the MAN_HO HR LAN. Two candidate locations were considered: the LDS Production1 LAN and the NCL Logistics LAN. The decision was based on performance modelling using provided link characteristics and anticipated workload requirements.

For this analysis, the maximum expected downloadable query result size was assumed to be 100 MB (approximately 800 Mb). It was further assumed that TCP is used for SQL data transfers and that links operate near their nominal bandwidth without persistent congestion. These assumptions are consistent with standard network performance modelling practices (Kurose & Ross, 2021).

The MAN_HO–LDS connection spans approximately 80 km and provides a bandwidth of 70 Mbps. Under ideal conditions, transmitting an 800 Mb payload over this link would require approximately 11.4 seconds, excluding protocol overhead **(Figure A)**. In contrast, the MAN_HO–NCL link, although longer at 220 km, provides a higher bandwidth of 100 Mbps, reducing the estimated transmission time to approximately 8 seconds **(Figure B)**. In both cases, propagation delay is negligible relative to transmission delay due to the comparatively short geographic distances involved.

Prior research has demonstrated that for bulk data transfers, available bandwidth has a significantly greater impact on end-to-end performance than propagation delay (Paxson &

Floyd, 1995). Furthermore, TCP throughput in wide-area networks is commonly constrained by link capacity rather than latency, particularly for large payloads (Jacobson, Braden & Borman, 1988).

Based on this performance analysis, the SQL server was placed within the NCL Logistics LAN. This choice minimises query response time for HR users while leveraging the higher-capacity WAN link within the MPLS VPN infrastructure. The placement therefore satisfies both performance efficiency and architectural requirements.

Option 1: SQL Server in LDS PRODUCTION 1

Distance: **80 km**
Bandwidth: **70 Mbps**
File size: **100 MB = 800 Mb (100 × 8 bits)**

Transmission time = 800 Mb ÷ 70 Mbps = 11.43 seconds

Propagation delay (assuming fiber optic):
Speed of light in fiber ≈ 200,000 km/s
Propagation delay = 80 km ÷ 200,000 km/s = 0.4 ms

Total time ≈ 11.43 seconds + 0.0004 seconds ≈ 11.43 seconds

**Figure A**

Option 2: SQL Server in NCL LOGISTICS **(Chosen)**

Distance: **220 km**
Bandwidth: **100 Mbps**
File size: **100 MB = 800 Mb**

Transmission time = 800 Mb ÷ 100 Mbps = 8 seconds

Propagation delay:
Propagation delay = 220 km ÷ 200,000 km/s = 1.1 ms

Total time ≈ 8 seconds + 0.0011 seconds ≈ 8 seconds

**Figure B**

## Requirement 5:

Requirement 5 will guarantee that NCL PRODUCTION2 LAN users have no access to MAN_HO EXEC LAN but have access to the rest of the network segments. This was done by including an augmented access control list to the router at the NCL Provider Edge closest PRODUCTION2. The access control policy blocks the PRODUCTION2 traffic and permits all other traffic to access the EXEC LAN. Verification testing works to make sure that PRODUCTION2 users can be allowed to access approved destinations and not the EXEC LAN or SQL server which validates the selective and effective access restriction.

## Requirement 6:

The requirement is based on creating an efficient and practical IP addressing scheme and making sure that client devices have their network configuration automatically set up through DHCP. To do so, Variable Length Subnet Masking (VLSM) was applied throughout the network wherein each LAN was given an address range that was assigned close to the actual size and usage of the LAN. This method prevents unnecessary large subnets and assists to reduce the wastage of the IP addresses, which is especially significant in IPv4-based enterprise networks (Okoh et al., 2024). This means that the network can be scaled and it can be easily maintained as it expands since each site only has the number of addresses it needs (Akintola and Falade, 2024).

Customer-edge (CE) routers were local DHCP servers in their respective LANs. Local DHCP services also make it simple to manage a network and minimize configuration errors by removing device-specific IP configuration. The IP allocation can be done locally even with the loss of WAN connectivity which enhances reliability. Congfinement of DHCP traffic in each site also does away with MPLS VPN broadcasts.

In an effort to provide a stable operation, infrastructure addresses such as default gateways were not allocated using DHCP. This will ensure that there will be no IP address clash between dynamically assigned client devices and the network equipment that is already configured. Consequently, clients are provided with valid IP addresses and correct subnet and gateway information, which provides instant and stable connectivity.

Overall, the combination of VLSM-based subnet design and distributed DHCP services provides a robust, scalable, and manageable addressing solution that satisfies both efficiency and operational requirements of the enterprise network.

| LAN | Network | Mask | Default Gateway | DHCP Range |
|---|---|---|---|---|
| **EXEC LAN** | 10.1.1.0 | /26 | 10.1.1.1 | 10.1.1.10–10.1.1.62 |
| **HR LAN** | 10.1.20.0 | /24 | 10.1.20.1 | 10.1.20.10–10.1.20.254 |

| | | | | |
|---|---|---|---|---|
| **R&D LAN A** | 10.3.1.0 | /26 | 10.3.1.1 | 10.3.1.10–10.3.1.62 |
| **R&D LAN B** | 10.3.2.0 | /28 | 10.3.2.1 | 10.3.2.10–10.3.2.14 |
| **LOGISTICS LAN** | 10.3.5.0 | /25 | 10.3.5.1 | 10.3.5.10–10.3.5.126 |
| **PRODUCTION2 LAN** | 10.3.3.0 | /23 | 10.3.3.1 | 10.3.3.10–10.3.4.254 |
| **PRODUCTION1 LAN** | 10.2.1.0 | /25 | 10.2.1.1 | 10.2.1.10–10.2.1.126 |

**Table 1**: LAN IP Addressing Scheme and DHCP Allocation

# SECTION 2

## Requirement 1: (Relevant router configuration commands)

Requirement 1 will mandate that the direct Ethernet connection between MAN_HO EXEC and HR should be secured because sensitive organisational data will pass over this connection. In order to fulfill this need, an site to site IPsec VPN was set up between the EXEC and HR routers to secure all the traffic over the line.

Both routers had the configuration done with mirrored settings and with a reciprocal address.

**Definition of Interesting Traffic**
Access control list (ACL) was set up to specify traffic that was to be secured by IPsec.

**access-list 101 permit ip host 10.1.12.1 host 10.1.12.2**

**Comment:**

- Specifies the traffic that will be encrypted by IPsec.
- Matches only traffic between the EXEC router (10.1.12.1) and the HR router (10.1.12.2).
- Ensures that only traffic on the direct EXEC–HR Ethernet link is protected, while all other traffic remains unaffected.

This ACL is a requirement that specifies that only traffic between the EXEC router (10.1.12.1) and the HR router (10.1.12.2) is to be encrypted. This will make sure that only sensitive inter-departmental traffic subjected to the direct Ethernet link will be put under IPsec protection

**IKE Phase 1 Configuration (ISAKMP Policy)**

Internet Key Exchange (IKE) Phase 1 was established so as to create a safe channel of control through which IPsec parameters will be negotiated.

**crypto isakmp policy 1**

**encryption aes**

**authentication pre-share**

**hash sha**

**Comment:**

- Configures a policy of the **ISAKMP** policy which was employed to define the secure IKE Phase 1 tunnel.
- **AES** is used to encrypt the IKE negotiation messages.
- **SHA** is implemented in order to assure integrity throughout authentication.
- **Pre-shared key** authentication is suitable for this controlled internal enterprise link.

**Pre-Shared Key Configuration**

Each router had a pre-shared key set to authenticate the remote peer in IKE Phase 1.

**crypto isakmp key KV6017SecureLink address 10.1.12.2**

**Comment:**

- Sets up a common secret to authentication of HR router.
- The peer IP address makes sure that the key is utilized to the actual remote device.
- The HR router is configured with the same key value that is referring to the IP address of the EXEC router.

**IKE Phase 2 Configuration (IPsec Transform Set)**

The IPsec Phase 2 parameters were set in such a way that the actual data traffic was secured.

**crypto ipsec transform-set EXEC-HR-VPN esp-aes esp-sha-hmac**

**Comment:**

- Defines the IPsec transform set used for data protection.
- AES offers secrecy, encryption of the payload.
- SHA-HMAC ensures data integrity and authentication.
- On both routers, there is a constant application of this transform set.

**Crypto Map Configuration**

A crypto map was constructed to tie up the peer, transform set as well as the interesting traffic and then used on the physical interface.

**crypto map EXEC-HR-CRYPTOMAP 10 ipsec-isakmp**

**set peer 10.1.12.2**

**set transform-set EXEC-HR-VPN**

**match address 101**

**Comment:**

- Relates the remote peer to the set IPsec parameters.
- Defines the transform set to be used in securing traffic.
- Associates the crypto map with the ACL that determines the encrypted traffic.
- The sequence number enables other VPNs to be added in case of necessity.

**Applying IPsec to the Interface**

The interface between EXEC and HR was filtered using the crypto map.

**interface FastEthernet2/0**

**crypto map EXEC-HR-CRYPTOMAP**

**Comment:**

- Switches on IPsec on the physical Ethernet interface.
- Ensures that all traffic matching the crypto ACL is encrypted before transmission.
- Traffic that does not correspond to the ACL is processed in a regular manner.

## Requirement 1 (Verification and Testing)

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **`show running-config** | section crypto` | Confirms that IPsec is configured on the EXEC router | The output indicates the configured ISAKMP policy, pre-shared key, IPsec transform set and crypto map in use on the interface. |

```
EXEC#show running-config | section crypto
crypto isakmp policy 1
 encr aes
 authentication pre-share
crypto isakmp key KV6017SecureLink address 10.1.12.2
crypto ipsec transform-set EXEC-HR-VPN esp-aes esp-sha-hmac
crypto map EXEC-HR-CRYPTOMAP 10 ipsec-isakmp
 set peer 10.1.12.2
 set transform-set EXEC-HR-VPN
 match address 101
 crypto map EXEC-HR-CRYPTOMAP
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show crypto isakmp policy** | Checks cryptographic parameters applied during IKE Phase 1 | The result confirms that AES encryption,SHA hashing, and pre-shared key authentication have been configured under ISAKMP policy 1. | Demonstrates that strong cryptographic mechanisms are used to establish a secure control channel between EXEC and HR |

```
EXEC#show crypto isakmp policy

Global IKE policy
Protection suite of priority 1
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #1 (768 bit)
        lifetime:               86400 seconds, no volume limit
Default protection suite
        encryption algorithm:   DES - Data Encryption Standard (56 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Rivest-Shamir-Adleman Signature
        Diffie-Hellman group:   #1 (768 bit)
        lifetime:               86400 seconds, no volume limit
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show access-lists 101** | Checks that the definition of interesting traffic is completed | The access-list specifically only allows EXEC-HR traffic to pass to be protected by IPsec only. | Confirms that only EXEC–HR traffic is selected for IPsec protection |

```
EXEC#show access-lists 101
Extended IP access list 101
    10 permit ip host 10.1.12.1 host 10.1.12.2 (76 matches)
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show crypto map** | set crypto map Verifies IPsec policy assembly/interface binding | The display includes the peer address, ACL association and transform set, and the interface crypto map | Confirms that IPsec parameters are correctly bound and applied to the EXEC–HR link |

```
EXEC#show crypto map
Crypto Map "EXEC-HR-CRYPTOMAP" 10 ipsec-isakmp
        Peer = 10.1.12.2
        Extended IP access list 101
            access-list 101 permit ip host 10.1.12.1 host 10.1.12.2
        Current peer: 10.1.12.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                EXEC-HR-VPN,
        }
        Interfaces using crypto map EXEC-HR-CRYPTOMAP:
                FastEthernet2/0
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
| --- | --- | --- | --- |
| **show running-config interface FastEthernet2/0** | Confirm the application of IPsec on the physical and interface | Checks running-config interface FastEthernet2/0 Checks whether the crypto map is set on the physical interface The result confirms that the crypto map is applied to the EXEC-HR Ethernet interface. | Confirms that all matching traffic on the link is encrypted |

```
EXEC#show running-config interface FastEthernet2/0
Building configuration...

Current configuration : 171 bytes
!
interface FastEthernet2/0
 description EXEC to HR - Secure Direct Link
 ip address 10.1.12.1 255.255.255.252
 duplex auto
 speed auto
 crypto map EXEC-HR-CRYPTOMAP
end

EXEC#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
| --- | --- | --- | --- |
| **show crypto isakmp sa** | Checks the IKE Phase 1 status | The QM_IDLE state means that the authentication and key exchange process is completed successfully between EXEC and HR. | Confirms that a secure IKE Phase 1 tunnel is established between EXEC and HR |

```
EXEC#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.1.12.2       10.1.12.1       QM_IDLE           1001    0 ACTIVE

IPv6 Crypto ISAKMP SA
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show crypto ipsec sa** | Checks the IPsec Phase 2 data protection | Packet encrypted and decrypted traffic is on the rise, which means that traffic between EXEC and HR is being encrypted. | Confirms that sensitive data transmitted between EXEC and HR is protected using IPsec |

```
EXEC#show crypto ipsec sa

interface: FastEthernet2/0
    Crypto map tag: EXEC-HR-CRYPTOMAP, local addr 10.1.12.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.1.12.1/255.255.255.255/0/0)
   remote ident (addr/mask/prot/port): (10.1.12.2/255.255.255.255/0/0)
   current_peer 10.1.12.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 20, #pkts encrypt: 20, #pkts digest: 20
    #pkts decaps: 20, #pkts decrypt: 20, #pkts verify: 20
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 5, #recv errors 0

     local crypto endpt.: 10.1.12.1, remote crypto endpt.: 10.1.12.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet2/0
     current outbound spi: 0x1A75510E(443896078)

     inbound esp sas:
      spi: 0x3376B7F9(863418361)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 3, flow_id: SW:3, crypto map: EXEC-HR-CRYPTOMAP
        sa timing: remaining key lifetime (k/sec): (4415590/2366)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0x1A75510E(443896078)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 4, flow_id: SW:4, crypto map: EXEC-HR-CRYPTOMAP
        sa timing: remaining key lifetime (k/sec): (4415590/2363)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE

     outbound ah sas:

     outbound pcp sas:
```

| **Verification Command** | **What the Command Verifies** | **Explanation of Output** | **How This Confirms the Requirement** |
|---|---|---|---|
| **show crypto session** | Checks IPsec tunnel and traffic flow | Verifies a status of an IPsec tunnel is UP-ACTIVE and shows the flow of traffic being correctly protected by the EXEC-HR IPsec tunnel. | Confirms that the EXEC–HR IPsec tunnel is active and correctly protecting defined traffic |

```
EXEC#show crypto session
Crypto session current status

Interface: FastEthernet2/0
Session status: UP-ACTIVE
Peer: 10.1.12.2 port 500
  IKE SA: local 10.1.12.1/500 remote 10.1.12.2/500 Active
  IPSEC FLOW: permit ip host 10.1.12.1 host 10.1.12.2
        Active SAs: 2, origin: crypto map

EXEC#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **traceroute 10.1.12.2** | Checks the route of the traffic between EXEC and HR | he trace route output reveals one hop straight to the HR router meaning that the traffic is going through direct Ethernet without any other router between it and the router. | Confirms that traffic follows the intended direct link while being protected by IPsec |

```
EXEC#traceroute 10.1.12.2

Type escape sequence to abort.
Tracing the route to 10.1.12.2

  1  *
    10.1.12.2 48 msec 76 msec
EXEC#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.1.12.2** | Checks encrypted connectivity | Successful ICMP responses can prove that the communication is functioning as well as encryption is active. | Demonstrates that secure communication does not disrupt connectivity between EXEC and H |

```
EXEC#ping 10.1.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/88/104 ms
```

```
EXEC#ping
Protocol [ip]:
Target IP address: 10.1.12.2
Repeat count [5]: 5
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 10.1.12.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.12.2, timeout is 2 seconds:
Packet sent with a source address of 10.1.12.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/83/88 ms
EXEC#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.1.12.1 (HR)** | Checks with encrypted bidirectional connectivity | ICMP response is successful which proves communication between HR and EXEC. | Ensures that the two-way communication is secure. |

```
HR#ping 10.1.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.12.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/85/92 ms
HR#
```

## Requirement 2 (Relevant router configuration commands)

To be clear and concise representative configurations of the MAN_HO PE router will be presented with the same logic to be applied to the LDS and NCL PE routers using site specific addressing.

**VRF Configuration (PE Routers)**

Each PE router was configured with a Virtual routing and forwarding (VRF) instance to provide isolation of customer routing information against the global routing table, and the exchange of routes among the participating sites on a controlled basis.

**VRF Creation**

**ip vrf CUSTOMER_A**

**rd 65000:100**

**route-target export 65000:100**

**route-target import 65000:100**

**Comments:**

- **Ip vrf CUSTOMER_A** command is used to generate a distinct routing table that is of the customer VPN.

```
MAN#show ip vrf
  Name                             Default RD          Interfaces
  CUSTOMER_A                       65000:100           Fa0/0
```

```
LDS#show ip vrf
  Name                             Default RD          Interfaces
  CUSTOMER_A                       65000:200           Fa0/1
LDS#
```

```
NCL#show ip vrf
  Name                             Default RD          Interfaces
  CUSTOMER_A                       65000:300           Fa0/0
```

- The Route Distinguisher **(RD)** is used to make the customer paths unique in the world of the MPLS network.
- Equivalent values of route-target **import,** and **export** values allow complete mesh route sharing amongst all PE routers involved in the CUSTOMER-A VPN.
- This design will enable the MAN HO HR, LD Production 1 and NCL Logistics to share routes safely and access networks without being connected to other VPNs.

**Customer-Facing Interface Assignment**

**interface FastEthernet0/0**

**ip vrf forwarding CUSTOMER_A**

**ip address 10.1.11.1 255.255.255.252**

**Comments:**

- The **ip vrf forwarding CUSTOMER-A** command allocates the customer facing interface to the VRF.
- This is necessary to make sure that all traffic passing into or out of the interface is handled using the CUSTOMER A routing table.
- The IP address is reconfigured once VRF assignment is done as dictated by Cisco IOS.
- Production 1 had the similar interface-to-VRF bindings configured on the LDS and NCL PE router respectively.

**MP-BGP Configuration (PE Routers)**

Each PE router was programmed to exchange VPNv4 routes with its peers in the MPLS core using MP-BGP. There was an establishment of internal BGP (iBGP) between all routers that were PE routers to provide complete mesh connectivity between client locations.

**BGP Process Initialisation**

**router bgp 65000**

**no bgp default ipv4-unicast**

**no auto-summary**

**no synchronization**

**Comments:**

- Starts the BGP with Autonomous System 65000.
- Disables automatic IPv4 unicast activation to allow explicit MP-BGP configuration.
- Eliminates route summarisation and IGP–BGP synchronization of legacy and route propagation is correct.

**iBGP Neighbour Configuration**

**neighbor 3.3.3.3 remote-as 65000**

**neighbor 3.3.3.3 update-source Loopback0**

**neighbor 4.4.4.4 remote-as 65000**

**neighbor 4.4.4.4 update-source Loopback0**

**Comments:**

- Establishes iBGP sessions with the LDS PE router (3.3.3.3) and the NCL PE router (4.4.4.4).
- All PE routers work in the same AS to facilitate MPLS VPN routing.
- Loopback interfaces are used as the update source to provide stable and resilient BGP sessions across the MPLS core.
- The LDS and NCL PE routers had equivalent neighbour configurations.

**VPNv4 Address Family Configuration**

**address-family vpnv4**

**neighbor 3.3.3.3 activate**

**neighbor 3.3.3.3 send-community extended**

**neighbor 4.4.4.4 activate**

**neighbor 4.4.4.4 send-community extended**

**exit-address-family**

**Comments:**

- Activates **VPNv4 address family** which is necessary to operate MPLS VPN.
- Enables sharing of VPNv4, between all PE routers.

- This is done by making sure that properly propagated route-target information is distributed by the **send-community extended** command to provide proper VRF route import and export.

- This network allows complete distribution of routes using the mesh between MAN_HO HR and the LLD Production 1 and NCL Logistics.

**VRF-Specific BGP Address Family**

**address-family ipv4 vrf CUSTOMER_A**

**redistribute connected**

**exit-address-family**

**Comments:**

- BGP configure of the **CUSTOMER_A VRF.**
- Transfers locally attached networks of customers into MP-BGP.
- Allows customer routes from each site to be advertised to all other PE routers, completing the full mesh VPN.

# Verification and Testing

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show mpls ldp neighbor** | Checks MPLS LDP neighbour relationships | The following indicates that the LDP neighbour table has been configured between the P router and all PE routers | Confirms that MPLS label distribution is operational across the core |

```
P#show mpls ldp neighbor
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
        TCP connection: 2.2.2.2.16752 - 1.1.1.1.646
        State: Oper; Msgs sent/rcvd: 279/283; Downstream
        Up time: 03:55:03
        LDP discovery sources:
          FastEthernet0/0, Src IP addr: 172.20.20.2
        Addresses bound to peer LDP Ident:
          10.1.10.1       172.20.20.2     2.2.2.2
    Peer LDP Ident: 4.4.4.4:0; Local LDP Ident 1.1.1.1:0
        TCP connection: 4.4.4.4.38858 - 1.1.1.1.646
        State: Oper; Msgs sent/rcvd: 281/284; Downstream
        Up time: 03:55:03
        LDP discovery sources:
          FastEthernet2/0, Src IP addr: 172.20.20.10
        Addresses bound to peer LDP Ident:
          10.3.10.1       172.20.20.10    4.4.4.4         10.3.11.1
    Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 1.1.1.1:0
        TCP connection: 3.3.3.3.64853 - 1.1.1.1.646
        State: Oper; Msgs sent/rcvd: 280/279; Downstream
        Up time: 03:54:49
        LDP discovery sources:
          FastEthernet0/1, Src IP addr: 172.20.20.6
        Addresses bound to peer LDP Ident:
          172.20.20.6     3.3.3.3
P#
```

```
MAN#show mpls ldp neighbor
    Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 2.2.2.2:0
        TCP connection: 1.1.1.1.646 - 2.2.2.2.30261
        State: Oper; Msgs sent/rcvd: 174/171; Downstream
        Up time: 02:21:13
        LDP discovery sources:
          FastEthernet2/0, Src IP addr: 172.20.20.1
        Addresses bound to peer LDP Ident:
          172.20.20.1     1.1.1.1         172.20.20.5     172.20.20.9
```

```
LDS#show mpls ldp neighbor
    Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 3.3.3.3:0
        TCP connection: 1.1.1.1.646 - 3.3.3.3.12540
        State: Oper; Msgs sent/rcvd: 170/173; Downstream
        Up time: 02:23:25
        LDP discovery sources:
          FastEthernet2/0, Src IP addr: 172.20.20.5
        Addresses bound to peer LDP Ident:
          172.20.20.1     1.1.1.1         172.20.20.5     172.20.20.9
LDS#
```

```
NCL#show mpls ldp neighbor
    Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 4.4.4.4:0
        TCP connection: 1.1.1.1.646 - 4.4.4.4.48189
        State: Oper; Msgs sent/rcvd: 175/174; Downstream
        Up time: 02:23:32
        LDP discovery sources:
          FastEthernet1/0, Src IP addr: 172.20.20.9
        Addresses bound to peer LDP Ident:
          172.20.20.1     1.1.1.1         172.20.20.5     172.20.20.9
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show mpls forwarding table** | Verifies MPLS label switching | The forwarding table shows local and remote labels used to forward traffic | Confirms that MPLS label-based forwarding is functioning correctly |

```
P#show mpls forwarding-table
Local  Outgoing    Prefix            Bytes tag  Outgoing   Next Hop
tag    tag or VC   or Tunnel Id      switched   interface
16     Pop tag     4.4.4.4/32        31197      Fa2/0      172.20.20.10
17     Pop tag     2.2.2.2/32        39460      Fa0/0      172.20.20.2
18     Pop tag     3.3.3.3/32        25433      Fa0/1      172.20.20.6
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip route vrf CUSTOMER_A** | Represents the routing table of the MPLS VPN VRF | The following output reveals the VRF routing table which includes one route to the local LAN network that is directly connected and several routes to remote locations learned by BGP. When there are BGP (B) routes, then MP-BGP VPNv4 is effectively distributing customer routes between sites within the same VRF. | Ensures that MAN Hr, LSD Production 1 and NCL logistics are linked in a complete mesh MPLS VPN and routing data is properly isolated inside the VRF. |

```
MAN#show ip route vrf CUSTOMER_A

Routing Table: CUSTOMER_A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.11.0 is directly connected, FastEthernet0/0
B       10.2.10.0 [200/0] via 3.3.3.3, 02:02:35
B       10.3.12.0 [200/0] via 4.4.4.4, 02:00:19
```

```
LDS#show ip route vrf CUSTOMER_A

Routing Table: CUSTOMER_A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
B       10.1.11.0 [200/0] via 2.2.2.2, 02:02:36
C       10.2.10.0 is directly connected, FastEthernet0/1
B       10.3.12.0 [200/0] via 4.4.4.4, 02:00:22
```

```
NCL#show ip route vrf CUSTOMER_A

Routing Table: CUSTOMER_A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
B       10.1.11.0 [200/0] via 2.2.2.2, 02:00:34
B       10.2.10.0 [200/0] via 3.3.3.3, 02:00:34
C       10.3.12.0 is directly connected, FastEthernet0/0
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip bgp vpnv4 vrf CUSTOMER_A** | Verifies MP-BGP VPNv4 route exchange | The result will show VPNv4 routes given and received on all sites. | Confirms that MP-BGP is distributing routes to support full mesh connectivity |

```
MAN#show ip bgp vpnv4 vrf CUSTOMER_A
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65000:100 (default for vrf CUSTOMER_A)
*> 10.1.11.0/30     0.0.0.0                  0         32768 ?
*>i10.2.10.0/30     3.3.3.3                  0    100      0 ?
*>i10.3.12.0/30     4.4.4.4                  0    100      0 ?
MAN#
```

```
LDS#show ip bgp vpnv4 vrf CUSTOMER_A
BGP table version is 7, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65000:200 (default for vrf CUSTOMER_A)
*>i10.1.11.0/30     2.2.2.2                  0    100      0 ?
*> 10.2.10.0/30     0.0.0.0                  0         32768 ?
*>i10.3.12.0/30     4.4.4.4                  0    100      0 ?
LDS#
```

```
NCL#show ip bgp vpnv4 vrf CUSTOMER_A
BGP table version is 7, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65000:300 (default for vrf CUSTOMER_A)
*>i10.1.11.0/30     2.2.2.2                  0    100      0 ?
*>i10.2.10.0/30     3.3.3.3                  0    100      0 ?
*> 10.3.12.0/30     0.0.0.0                  0         32768 ?
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping vrf CUSTOMER_A <remote_LAN_IP>** | Checks the end-to-end connectivity | Succeeding ICMP responses confirm the connectivity between the sites | Checks to verify that the entire mesh MPLS VPN is operational. |

```
MAN#ping vrf CUSTOMER_A 10.2.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/89/108 ms
```

```
MAN#ping vrf CUSTOMER_A 10.3.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/87/92 ms
```

## Requirement 3: (Relevant router configuration commands)

Requirement 3 requires that only the sensitive traffic between the EXEC LAN and R&D LAN-A should be encrypted and the rest of the traffic between the CUSTOMER B should be passed through MPLS Layer 3 VPN without encryption.

### MAN_HO (Provider Edge Router - Manchester Head Office)

**VRF-Configuration of Direct VPN.**

**ip vrf CUSTOMER_B**

description Direct VPN EXEC to R&D (Requirement 3)

**rd 65000:400**

**route-target export 65000:200**

**route-target import 65000:200**

**Comment:**

- Make **VRF CUSTOMER-B** to separate Requirement 3 traffic with that of other VPNs
- **RD 65000:400:** Route Distinguishes the routes throughout the world in MP-BGP.
- The VPN membership is under control of route target **(RT) 65000:200**.
- In both import and export, the RT value is the same to form a direct VPN with NCL.

**Configuration of Customer-Facing Interface.**

**interface FastEthernet0/1**

description Customer connection to EXC router

**ip vrf forwarding CUSTOMER_B**

```
 ip address 10.1.10.1 255.255.255.252
 no shutdown
```

**Comment:**

- The associates contact **VRF CUSTOMER_B** at the point of isolating traffic.
- Relates to EXEC CE router with/30 point to point subnet.
- **VRF forwarding** is used to maintain separation of EXEC traffic among other customers.

**BGP Settings to Exchange VPNv4 Routes.**

```
router bgp 65000
 address-family ipv4 vrf CUSTOMER_B
  redistribute connected
  redistribute static
  no synchronization
 exit-address-family
```

**Comment:**

- Allows MP-BGP to share VPNv4 between PE routers.
- Re-marks outright linkage customer interface into BGP.
- provides NCL with a redistributed route (EXEC LAN) to BGP.
- This enables automatic exchange of routes between MAN_HO and NCL.

**Static Route to EXEC LAN**

```
ip route vrf CUSTOMER_B 10.1.1.0 255.255.255.192 10.1.10.2
```

**Comment:**

- Defines how to reach EXEC LAN (10.1.1.0/26) via EXEC CE router (10.1.10.2).
- This path is re-employed into MP-BGP and is announced to NCL PE router.
- Helps NCL to know the route to the network of EXEC.

## NCL (Provider Edge Router - Newcastle)

**VRF Configuration (MUST match route-target with MAN_HO)**

**ip vrf CUSTOMER_B**

description Direct VPN EXEC to R&D (Requirement 3)

**rd 65000:500**

**route-target export 65000:200**

**route-target import 65000:200**

**Comment:**

- Different RD (65000:500) but SAME route-target (65000:200) creates direct VPN.
- Matching RTvalues allow automatic exchange of routes with MAN HO.
- RT 65000:200 routes are imported into this VRF by MAN_HO.

**Configuration of Customer-Facing Interface.**

**interface FastEthernet0/1**

description Customer connection to R & D router.

**ip vrf forwarding CUSTOMER_B**

**ip address 10.3.10.1 255.255.255.252**

**no shutdown**

**Comment:**

- connects to RND CE router with /30 point-to-point subnet.
- VRF CUSTOMER B segregates traffic on R&D customers.

**BGP Configuration**

**router bgp 65000**

**address-family ipv4 vrf CUSTOMER_B**

**redistribute connected**

**redistribute static**

**no synchronization**

**exit-address-family**

**Comment:**

• MP-BGP routes with CUSTOMER-B VPN to MAN-HO.

• Advertises R&D networks to MAN_HO so EXEC can reach them.

**Statics to R&D LANs (LAN A and LAN B)**

**ip route vrf CUSTOMER_B 10.3.1.0 255.255.255.192 10.3.10.2**

**ip route vrf CUSTOMER_B 10.3.2.0 255.255.255.240 10.3.10.2**

**Comment:**

• The first route: 10.3.1.0/26 R&D LAN A with 50 users will be encrypted using IPsec.

• Second pathway: R and D LAN B (10.3.2.0/28) - 12 users will not be encrypted.

• Both routes via RND CE router (10.3.10.2).

• Both paths are announced to MAN-HO through MP-BGP to EXEC so as to access them.

## EXEC (Customer Edge Router)

**IKE Phase 1 Policy (ISAKMP)**

**crypto isakmp policy 20**

**encryption aes**

**hash sha**

**authentication pre-share**

**group 2**

**lifetime 86400**

**Comment:**

• ISAKMP policy setting to establish IKE Phase 1 tunnel.

• AES-128 encryption involves sufficient security and fewer computational resources.

• SHA-1 authentication - appropriate to in-house VPN.

• Enterprise internal linkage Pre-shared key authentication.

• Diffie-Hellman Group 2 (1024-bit) offers the balance between the security and performance.

• 24-hour (86400 seconds) Security Association lifetime.

**Pre-Shared Key for IPsec Peer Authentication**

```
crypto isakmp key Cisco123 address 10.3.10.2
```

**Comment:**

• Sets up a common secret, Cisco123 to authenticate RND router.

• The IP address of the WAN of RND is 10.3.10.2.

• Key should be a perfect match on RND router (case-sensitive).

• This key identifies the remote peer in the negotiation in the IKE Phase 1

**IPsec Transform Set (Phase 2)**

```
crypto ipsec transform-set REQ3-TS esp-aes esp-sha-hmac
 mode tunnel
```

**Comment:**

• Identifies encryption and authentication data protection algorithms.

• ESP (Encapsulating Security Payload) with AES encryption for confidentiality.

• SHA HMAC offers integrity and authentication of data.

• The IP packet is fully encapsulated by tunnel mode and is fully protected.

**Access Control List - Selective Encryption (ONLY LAN A)**

```
ip access-list extended REQ3-LAN-A
 permit ip 10.1.1.0 0.0.0.63 10.3.1.0 0.0.0.63
```

**Comment:**

• Determines interesting traffic to IPsec - traffic to be encrypted.

• Source: 10.1.1.0/26 (EXEC LAN - 50 users).

• Destination: 10.3.1.0/26 (R&D LAN A - 50 users).

• CRITICAL: LAN B (10.3.2.0/28) is NOT included in this ACL.

• Traffic sent in LAN A is the only one which matches this ACL and will be encrypted by IPsec.

• LAN B Traffic will be sent through the MPLS VPN without IPsec encryption overhead.

• This shows how low-sensitivity data has performance-security trade-off.

**Crypto Map (binds policy, peer, and ACL)**

```
crypto map EXEC-RND-MAP 20 ipsec-isakmp
 set peer 10.3.10.2
 set transform-set REQ3-TS
 match address REQ3-LAN-A
```

**Comment:**

• Associates IPsec parameters for the VPN tunnel.

• The WAN interface of the RND router (remote IPsec endpoint) is named Peer 10.3.10.2.

• Transform set REQ3-TS encodes algorithms of encryption/authentication.

• ACL REQ3-LAN-A determines which traffic gets encrypted (only LAN A).

• Crypto map 20 permits extra crypto maps where necessary.

**Apply Crypto Map to WAN Interface**

```
interface FastEthernet0/0
 description Link to MAN_HO
 ip address 10.1.10.2 255.255.255.252
 crypto map EXEC-RND-MAP
 no shutdown
```

**Comment:**

• Enables IPsec policy on the interface that is facing the MPLS VPN cloud.

• Traffic that is equivalent to ACL REQ3-LAN-A will be encrypted and then sent.

• Traffic not equal to ACL is sent through without undergoing IPsec.

• Encrypted IPsec packets are encapsulated with MPLS labels by MAN_HO PE router.

## RND (Customer Edge Router)

**IKE Phase 1 Policy (MUST MATCH EXEC exactly)**

**crypto isakmp policy 20**

**encryption aes**

**hash sha**

**authentication pre-share**

**group 2**

**lifetime 86400**

**Comment:**

- Same ISAKMP policy as EXEC router to allow Phase 1 negotiation to be successful.
- Tunnel failure due to any mismatch in encryption, hash or DH group will occur.
- There should be a mutual matching of policy between peers in order to select proposals.

**Pre-Shared Key (SAME as EXEC, pointing to EXEC's IP)**

**crypto isakmp key Cisco123 address 10.1.10.2**

**Comment:**

- Shared secret "Cisco123" must match EXEC router exactly.
- Peer address 10.1.10.2 is the IP address of the WAN interface of EXEC.
- Symmetric authentication - both the routers share a common key.

**IPsec Transform Set (MUST MATCH EXEC)**

**crypto ipsec transform-set REQ3-TS esp-aes esp-sha-hmac**

**mode tunnel**

**Comment:**

- Same name of transform set and algorithms as EXEC router.

• Ensures both peers can successfully negotiate Phase 2 parameters.

**Access Control List - Mirror of EXEC (source/dest reversed)**

**ip access-list extended REQ3-LAN-A**

**description Encrypt only RND LAN A traffic to EXEC**

**permit ip 10.3.1.0 0.0.0.63 10.1.1.0 0.0.0.63**

**Comment:**

- **ip access-list extended command, This command is used to define an interest extending ACL named with the ip access-list extended command.**
- The permit ip statement tells that the packets between R&D LAN A (10.3.1.0/26) and the EXEC LAN (10.1.1.0/26) can be encrypted by IPsec
- The crypto map uses this ACL to identify traffic that is secured by IPsec.
- Any traffic that is not equal to this ACL is passed on to the MPLS VPN in a regular manner, and not encrypted, such as R&D LAN B (10.3.2.0/28).
- The ACL mirrors the EXEC-side ACL with source and destination reversed to ensure bidirectional IPsec operation.

**Crypto Map**

**crypto map EXEC-RND-MAP 20 ipsec-isakmp**

**description IPSec to EXEC for LAN A only**

**set peer 10.1.10.2**

**set transform-set REQ3-TS**

**match address REQ3-LAN-A**

**Comment:**

• The WAN interface of EXEC router is 10.1.10.2.

• IPsec-policy Set and ACL are linked together using links.

• There will be only a traffic matching REQ3-LAN-A ACL that will be encrypted.

**Apply to WAN Interface**

**interface FastEthernet0/0**

**description Link to NCL PE Router**

**ip address 10.3.10.2 255.255.255.252**

**crypto map EXEC-RND-MAP**

**no shutdown**

**Comment:**

• switches on IPsec on interface to NCL PE router.

• LAN A traffic is encrypted prior to the entry of MPLS VPN cloud.

• LAN B traffic is not processed in IPsec.

## Verification and Testing

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip route vrf CUSTOMER_B** | Make sure routing within the MPLS VPN | Output shows EXEC LAN (10.1.1.0/26) and R&D LANs (10.3.1.0/26, 10.3.2.0/28) reachable via CUSTOMER_B | enables verification of direct MPLS Layer 3 VPN between MAN_HO EXEC and NCL R&D. |

```
MAN#show ip route vrf CUSTOMER_B

Routing Table: CUSTOMER_B
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
C       10.1.10.0/30 is directly connected, FastEthernet0/1
B       10.3.10.0/30 [200/0] via 4.4.4.4, 13:57:41
B       10.3.1.0/26 [200/0] via 4.4.4.4, 14:37:18
S       10.1.1.0/26 [1/0] via 10.1.10.2
B       10.3.2.0/28 [200/0] via 4.4.4.4, 14:37:18
MAN#
```

```
NCL#show ip route vrf CUSTOMER_B

Routing Table: CUSTOMER_B
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
B       10.1.10.0/30 [200/0] via 2.2.2.2, 14:10:14
C       10.3.10.0/30 is directly connected, FastEthernet0/1
S       10.3.1.0/26 [1/0] via 10.3.10.2
B       10.1.1.0/26 [200/0] via 2.2.2.2, 11:21:51
S       10.3.2.0/28 [1/0] via 10.3.10.2
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip bgp vpnv4 vrf CUSTOMER_B** | Verifies MP-BGP route distribution | VPNv4s have proper RD and next-hop via MPLS core. | Ensures that the correct route exchange between the EXEC-R&D VPN is with the direct EXEC. |

```
MAN#show ip bgp vpnv4 vrf CUSTOMER_B
BGP table version is 67, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65000:400 (default for vrf CUSTOMER_B)
*> 10.1.1.0/26      10.1.10.2                0         32768 ?
*> 10.1.10.0/30     0.0.0.0                  0         32768 ?
*>i10.3.1.0/26      4.4.4.4                  0    100      0 ?
*>i10.3.2.0/28      4.4.4.4                  0    100      0 ?
*>i10.3.10.0/30     4.4.4.4                  0    100      0 ?
MAN#
```

```
NCL#show ip bgp vpnv4 vrf CUSTOMER_B
BGP table version is 65, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65000:500 (default for vrf CUSTOMER_B)
*>i10.1.1.0/26      2.2.2.2                  0    100      0 ?
*>i10.1.10.0/30     2.2.2.2                  0    100      0 ?
*> 10.3.1.0/26      10.3.10.2                0         32768 ?
*> 10.3.2.0/28      10.3.10.2                0         32768 ?
*> 10.3.10.0/30     0.0.0.0                  0         32768 ?
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping vrf CUSTOMER_B 10.3.1.1/ 10.3.2.1** | Checks reachability to R&D LAN A/B | ICMP replies are received successfully | Checks reachability of the direct MPLS VPN; LAN A/B. |

```
MAN#ping vrf CUSTOMER_B 10.3.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/84/108 ms
MAN#ping vrf CUSTOMER_B 10.3.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/86/96 ms
```

**In NCL for EXEC LAN:**

```
NCL#ping vrf CUSTOMER_B  10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/96/108 ms
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **traceroute vrf CUSTOMER_B 10.3.1.1** | Checks traffic route | Trace demonstrates traffic using MPLS VPN direct route | Checks that traffic is using direct VPN route. |

```
MAN#traceroute vrf CUSTOMER_B 10.3.1.1

Type escape sequence to abort.
Tracing the route to 10.3.1.1

  1 172.20.20.1 [MPLS: Labels 17/23 Exp 0] 68 msec 76 msec 92 msec
  2 10.3.10.1 [MPLS: Label 23 Exp 0] 36 msec 84 msec 60 msec
  3 10.3.10.2 76 msec 104 msec 76 msec
MAN#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show crypto isakmp sa** | Checks IKE Phase 1 tunnel | The state QM_IDLE indicates successful negotiation | indicates that a secure IKE control channel has been set up between EXEC and R&D. |

```
EXEC#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.3.10.2       10.1.10.2       QM_IDLE           1004    0 ACTIVE
10.1.12.2       10.1.12.1       QM_IDLE           1001    0 ACTIVE

IPv6 Crypto ISAKMP SA
```

```
RND#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.3.10.2       10.1.10.2       QM_IDLE           1003    0 ACTIVE

IPv6 Crypto ISAKMP SA
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.3.1.1 source 10.1.1.1** | Tests encrypted traffic | Traffic matches IPsec ACL and is encrypted | Confirms selective encryption of LAN A |
| **show crypto ipsec sa** | Checks that IPsec Phase 2 encryption | Security Associations are present in EXEC LAN - R&D LAN A and packet counts are rising. | Confirms that **LAN A traffic is encrypted using IPsec** |

```
EXEC#ping 10.3.1.1 source 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 132/175/228 ms
EXEC#show crypto ipsec sa

interface: FastEthernet2/0
    Crypto map tag: EXEC-HR-CRYPTOMAP, local addr 10.1.12.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.1.12.1/255.255.255.255/0/0)
   remote ident (addr/mask/prot/port): (10.1.12.2/255.255.255.255/0/0)
   current_peer 10.1.12.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.1.12.1, remote crypto endpt.: 10.1.12.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet2/0
     current outbound spi: 0x0(0)

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:

interface: FastEthernet0/0
    Crypto map tag: EXEC-RND-MAP, local addr 10.1.10.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.1.1.0/255.255.255.192/0/0)
   remote ident (addr/mask/prot/port): (10.3.1.0/255.255.255.192/0/0)
   current_peer 10.3.10.2 port 500
     PERMIT, flags={origin_is_acl,ipsec_sa_request_sent}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
```

```
RND#ping 10.1.1.1 source 10.3.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.3.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 124/193/252 ms
RND#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: EXEC-RND-MAP, local addr 10.3.10.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.3.1.0/255.255.255.192/0/0)
   remote ident (addr/mask/prot/port): (10.1.1.0/255.255.255.192/0/0)
   current_peer 10.1.10.2 port 500
     PERMIT, flags={origin_is_acl,ipsec_sa_request_sent}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

     local crypto endpt.: 10.3.10.2, remote crypto endpt.: 10.1.10.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x30CB3C10(818625552)

     inbound esp sas:
      spi: 0xEFCD54EB(4023211243)
        transform: esp-aes esp-sha-hmac ,
 --More--
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.3.2.1 source 10.1.1.1** | Tests unencrypted traffic | Traffic passes without IPsec SA | IPsec confirm LAN B bypasses IPsec |

```
EXEC#ping 10.3.2.1 source 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 160/226/308 ms
```

```
RND#ping 10.1.1.1 source 10.3.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.3.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 152/181/220 ms
RND#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: EXEC-RND-MAP, local addr 10.3.10.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.3.1.0/255.255.255.192/0/0)
   remote ident (addr/mask/prot/port): (10.1.1.0/255.255.255.192/0/0)
   current_peer 10.1.10.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 121, #pkts encrypt: 121, #pkts digest: 121
    #pkts decaps: 121, #pkts decrypt: 121, #pkts verify: 121
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.3.10.2, remote crypto endpt.: 10.1.10.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:
RND#
```

```
EXEC#show crypto ipsec sa | include 10.3.2.0
EXEC#
```

## Requirement 4: (Relevant router configuration commands)

In order to control access to the company-internal SQL server, representative structures of the LOGISTICS router are provided. The same logical approach applies across the MPLS VPN, where access control is implemented at the nearest edge to the resource being protected.

**SQL Server Network Configuration (LOGISTICS Router)**

The LOGISTICS router was set up to have the SQL server as part of its local LAN and give it access to the MPLS VPN through the NCL PE router.

**LOGISTICS LAN Interface Configuration**

**interface FastEthernet0/0**

**description LOGISTICS LAN (SQL Server)**

**ip address 10.3.5.1 255.255.255.128**

**no shutdown**

**Comments:**

- This interface provides Layer-3 connectivity to the LOGISTICS LAN where the SQL server is located.
- The SQL server subnet is the default gateway of the IP address.
- Subnet of /25 enables adequate addressing capacity whilst being well sized.

**WAN Interface Configuration (Toward NCL PE)**

**interface FastEthernet0/1**

**description Link to NCL PE Router**

**ip address 10.3.12.2 255.255.255.252**

**no shutdown**

**Comments:**

- This interface links the LOGISTICS site and the MPLS VPN through the NCL PE router.
- /30 subnet is employed to offer a point to point WAN connection with minimal address wastage.
- All the traffic between MAN- HO HR and SQL server passes through this interface.

**Default Route Configuration**

**ip route 0.0.0.0 0.0.0.0 10.3.12.1**

**Comments:**

- All non-local traffic is directed to the MPLS VPN by a default route.
- This makes sure that there is reachability between LOGISTICS and remote customer sites that are part of the VPN.
- The destination address is that of the NCL PE router.

**Access Control Configuration (HR-Only SQL Access)**

In order to implement the requirement of ensuring that only MAN HR users can access the SQL server, an extended access control list was added to the LOGISTICS router.

**Extended ACL Definition**

```
ip access-list extended HR-ONLY-SQL
 remark Allow HR LAN access to SQL server
 permit ip 10.1.20.0 0.0.0.255 host 10.3.5.1

 remark Deny all other access to SQL server
 deny ip any host 10.3.5.1 log

 remark Allow all other traffic
 permit ip any any
```

**Comments:**

- The first rule explicitly permits traffic from the MAN_HO HR LAN to the SQL server.
- The second rule blocks access attempts of all other sources and implements the least privilege principle.
- The logging is also turned on to be able to see unauthorised access attempts.
- The final permit statement helps to avoid unintentional interference with the traffic that is not related.

**ACL Application to WAN Interface**

```
interface FastEthernet0/1
 ip access-group HR-ONLY-SQL in
```

**Comments:**

- The ACL is applied on the inbound on the WAN interface in order to block incoming traffic of the MPLS network.
- This location makes access control imposed as near as possible to the SQL server.
- Filtering at the LOGISTICS router can be centralised to make it easy to manage and enhance security.

## Verification and Testing

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip route** | Checks the availability of local and remote routes | Checks the connected LOGISTICS LAN routes, and the default route leading to the MPLS network | Confirms correct routing toward HR via the MPLS VPN |

```
LOGISTICS#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.3.12.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C       10.3.12.0/30 is directly connected, FastEthernet0/1
C       10.3.5.0/25 is directly connected, FastEthernet0/0
S       10.1.20.0/24 [1/0] via 10.3.12.1
S*   0.0.0.0/0 [1/0] via 10.3.12.1
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.3.5.1 source 10.1.20.1** | Checks access by HR to SQL server | Successful replies of ICMP mean authorized access to SQL server. | Confirms HR users can reach the SQL server |

```
HR#ping 10.3.5.1 source 10.1.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.5.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.20.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 188/218/280 ms
HR#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.3.5.1 source <non-HR-IP>** | Tests unauthorised access | ICMP requests fail | Checks that access is restricted to HR only. |

```
EXEC#ping 10.3.5.1 source 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.5.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
UUUUU
Success rate is 0 percent (0/5)
EXEC#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show access-lists HR-ONLY-SQL** | Checks operation of ACLs | hit counts go up on permit and deny records | Checks ACL enforcement is working properly |

```
LOGISTICS#show access-lists HR-ONLY-SQL
Extended IP access list HR-ONLY-SQL
    10 permit ip 10.1.20.0 0.0.0.255 host 10.3.5.1 (30 matches)
    20 deny ip any host 10.3.5.1 log
    30 permit ip any any
```

## Requirement 5: (Relevant router configuration commands)

**Access Control Configuration (NCL PE Router)**

**ip access-list extended BLOCK-PROD2-EXEC**

**deny ip 10.3.2.0 0.0.1.255 10.1.1.0 0.0.0.63**

**permit ip any any**

**Comments:**

- The initial rule specifically prevents traffic in the PRODUCTION2 LAN, going to the EXEC LAN.
- The wildcard mask covers the entire PRODUCTION2 subnet.
- The last permit statement treats all other legitimate traffic that is not to be disrupted accidentally.

**ACL Application to PRODUCTION2 Link**

**interface FastEthernet2/0**

**description Link to PRODUCTION2 - CUSTOMER_A**

**ip vrf forwarding CUSTOMER_A**

**ip address 10.3.11.1 255.255.255.252**

**ip access-group BLOCK-PROD2-EXEC in**

**Comments:**

- The ACL is implemented inbound on the interface between PRODUCTION2 and MPLS VPN.
- This location imposes access control as near to the traffic source as possible.

- Using the ACL at the PE router will prevent the avoidable traffic flow over the MPLS core.

## Verification and Testing

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show running-config interface FastEthernet2/0** | Confirms ACL placement | Confirms that we have placed ACLs inbound on PRODUCTION2-facing interface | Confirm that we have placed ACLs in the correct location |

```
NCL#show running-config interface FastEthernet2/0
Building configuration...

Current configuration : 209 bytes
!
interface FastEthernet2/0
 description Link to PRODUCTION2 - CUSTOMER_A
 ip vrf forwarding CUSTOMER_A
 ip address 10.3.11.1 255.255.255.252
 ip access-group BLOCK-PROD2-EXEC in
 duplex auto
 speed auto
end

NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show access-lists BLOCK-PROD2-EXEC** | Checks ACL rules and activity | Reports deny and permit rules with hits | check EXEC traffic is blocked explicitly |

```
NCL#show access-lists BLOCK-PROD2-EXEC
Extended IP access list BLOCK-PROD2-EXEC
    10 deny ip 10.3.2.0 0.0.1.255 10.1.1.0 0.0.0.63
    20 permit ip any any (104 matches)
NCL#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.1.20.1 (HR)** | Tests permitted access | successful ICMP response | Confirms that PRODUCTION2 can get to permitted LANs. |

```
PRODUCTION2#ping 10.1.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 188/228/292 ms
PRODUCTION2#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.1.1.1(EXEC)** | Tests denied access to EXEC | ICMP requests fail | Confirms EXEC LAN is protected |

```
PRODUCTION2#ping 10.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
PRODUCTION2#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **traceroute 10.1.1.1** | Identifies blocking point | Traffic is sent to PE and dropped at that point | Confirms ACL-based denial |

```
PRODUCTION2#traceroute 10.1.1.1

Type escape sequence to abort.
Tracing the route to 10.1.1.1

  1 10.3.11.1 100 msec 92 msec 112 msec
  2 10.3.11.1 !H  !H  !H
PRODUCTION2#
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.3.5.1** | Tests access to SQL server | ICMP denied by HR-only ACL | Tests that layered security policies have been maintained. |

```
PRODUCTION2#ping 10.3.5.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.5.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
PRODUCTION2#
```

## Requirement 6: (Relevant router configuration commands)

To implement dynamic IP address allocation, DHCP services were configured **on each CE router.** The following configuration demonstrates a representative example; the same logical approach was applied to all remaining CE routers using site-specific addressing.

**DHCP Excluded Address Configuration (EXEC)**

**ip dhcp excluded-address 10.1.1.1 10.1.1.9**

**Comments:**
•Reserves default gateway and low number infrastructure addresses.
• Stops the assignment of addresses which has been configured as static on network machines by DHCP.

• Minimizes the possibility of IP address conflict and enhances reliability.

**DHCP Pool Configuration**

**ip dhcp pool EXEC_LAN**

**network 10.1.1.0 255.255.255.192**

**default-router 10.1.1.1**

**Comments:**
• Defines the subnet from which client IP addresses are dynamically allocated.
• Default-router command can be used so that all the clients are provided with the proper gateway address.
• A /26 subnet will offer enough capacity and at the same time efficient utilisation of the address.

## Verification and Testing

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip dhcp pool** | Verifies DHCP pool and configuration | Displays the configured DHCP pool and available address range | Verifies that DHCP is properly configured and active |

```
EXEC#show ip dhcp pool

Pool EXEC_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.1.1.3             10.1.1.1         - 10.1.1.62         1
```

**More testing verification:**

**MAN**

HR

```
HR#show ip dhcp pool

Pool EXEC_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.1.20.3            10.1.20.1        - 10.1.20.254       1
HR#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.1.20.2           0100.5079.6668.03       Mar 03 2002 08:31 PM    Automatic
```

**NCL**

## R&D

```
RND#show ip dhcp pool

Pool R&D_LANA :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.3.1.3             10.3.1.1         - 10.3.1.62         1

Pool R&D_LANB :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 14
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.3.2.3             10.3.2.1         - 10.3.2.14         1
RND#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.3.1.2            0100.5079.6668.01       Mar 03 2002 09:38 PM    Automatic
10.3.2.2            0100.5079.6668.00       Mar 03 2002 09:42 PM    Automatic
RND#
```

LAN A (PC)

```
LANA> ip dhcp
DDORA IP 10.3.1.2/26 GW 10.3.1.1

LANA> show

NAME   IP/MASK              GATEWAY           MAC                LPORT  RHOST:PORT
LANA   10.3.1.2/26          10.3.1.1          00:50:79:66:68:01  10070  127.0.0.1:10071
       fe80::250:79ff:fe66:6801/64

LANA> ping
LANA> ping  10.3.1.1
84 bytes from 10.3.1.1 icmp_seq=1 ttl=255 time=47.270 ms
84 bytes from 10.3.1.1 icmp_seq=2 ttl=255 time=15.999 ms
84 bytes from 10.3.1.1 icmp_seq=3 ttl=255 time=15.093 ms
84 bytes from 10.3.1.1 icmp_seq=4 ttl=255 time=15.498 ms
84 bytes from 10.3.1.1 icmp_seq=5 ttl=255 time=16.339 ms

LANA>
```

LAN B (PC)

```
LANB> ip dhcp
DORA IP 10.3.2.2/28 GW 10.3.2.1

LANB> show

NAME   IP/MASK              GATEWAY           MAC                LPORT  RHOST:PORT
LANB   10.3.2.2/28          10.3.2.1          00:50:79:66:68:00  10068  127.0.0.1:10069
       fe80::250:79ff:fe66:6800/64

LANB> ping 10.3.2.1
84 bytes from 10.3.2.1 icmp_seq=1 ttl=255 time=15.507 ms
84 bytes from 10.3.2.1 icmp_seq=2 ttl=255 time=17.708 ms
84 bytes from 10.3.2.1 icmp_seq=3 ttl=255 time=14.542 ms
84 bytes from 10.3.2.1 icmp_seq=4 ttl=255 time=11.516 ms
84 bytes from 10.3.2.1 icmp_seq=5 ttl=255 time=15.034 ms

LANB>
```

## PRODUCTION 2

```
PRODUCTION2#show ip dhcp pool

Pool Production2_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.3.2.2             10.3.2.1         - 10.3.3.254        1
PRODUCTION2#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.3.2.1            0100.5079.6668.05       Mar 03 2002 07:44 AM    Automatic
PRODUCTION2#
```

```
PC1> IP 10.3.2.1/23 GW 10.3.3.1
DORA
PC1> show

NAME   IP/MASK              GATEWAY           MAC                LPORT  RHOST:PORT
PC1    10.3.2.1/23          10.3.3.1          00:50:79:66:68:05  10094  127.0.0.1:10095
       fe80::250:79ff:fe66:6805/64

PC1> ping 10.3.3.1
84 bytes from 10.3.3.1 icmp_seq=1 ttl=255 time=15.595 ms
84 bytes from 10.3.3.1 icmp_seq=2 ttl=255 time=16.112 ms
84 bytes from 10.3.3.1 icmp_seq=3 ttl=255 time=15.328 ms
84 bytes from 10.3.3.1 icmp_seq=4 ttl=255 time=14.982 ms
84 bytes from 10.3.3.1 icmp_seq=5 ttl=255 time=14.360 ms

PC1>
```

LOGISTICS

```
LOGISTICS#show ip dhcp pool

Pool Logistics_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.3.5.3             10.3.5.1         - 10.3.5.126        1
LOGISTICS#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.3.5.2            0100.5079.6668.04       Mar 03 2002 09:58 AM    Automatic
```

```
PC1> ip dhcp
DDORA IP 10.3.5.2/25 GW 10.3.5.1

PC1> show

NAME   IP/MASK              GATEWAY          MAC                LPORT  RHOST:PORT
PC1    10.3.5.2/25          10.3.5.1         00:50:79:66:68:04  10088  127.0.0.1:10089
       fe80::250:79ff:fe66:6804/64

PC1> ping
PC1> ping 10.3.5.1
84 bytes from 10.3.5.1 icmp_seq=1 ttl=255 time=17.773 ms
84 bytes from 10.3.5.1 icmp_seq=2 ttl=255 time=16.493 ms
84 bytes from 10.3.5.1 icmp_seq=3 ttl=255 time=14.983 ms
84 bytes from 10.3.5.1 icmp_seq=4 ttl=255 time=14.785 ms
84 bytes from 10.3.5.1 icmp_seq=5 ttl=255 time=15.039 ms
```

**LDS**

PRODUCTION 1

```
PRODUCTION1#show ip dhcp pool

Pool PRODUCTION1_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.2.1.3             10.2.1.1         - 10.2.1.126        1
PRODUCTION1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.2.1.2            0100.5079.6668.06       Mar 03 2002 09:04 AM    Automatic
PRODUCTION1#
```

```
PC1> ip dhcp
DORA IP 10.2.1.2/25 GW 10.2.1.1

PC1> show

NAME   IP/MASK              GATEWAY          MAC                LPORT  RHOST:PORT
PC1    10.2.1.2/25          10.2.1.1         00:50:79:66:68:06  10100  127.0.0.1:10101
       fe80::250:79ff:fe66:6806/64

PC1> ping 10.2.1.1
84 bytes from 10.2.1.1 icmp_seq=1 ttl=255 time=14.856 ms
84 bytes from 10.2.1.1 icmp_seq=2 ttl=255 time=15.919 ms
84 bytes from 10.2.1.1 icmp_seq=3 ttl=255 time=15.364 ms
84 bytes from 10.2.1.1 icmp_seq=4 ttl=255 time=15.643 ms
84 bytes from 10.2.1.1 icmp_seq=5 ttl=255 time=15.066 ms
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **show ip dhcp binding** | checks on DHCP lease assignments | Shows dynamically assigned IP addresses mapped to client MAC addresses | Confirms that clients receive IP addresses via DHCP |

```
EXEC#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
10.1.1.2            0100.5079.6668.02       Mar 03 2002 10:19 PM    Automatic
```

| Verification Command | What the Command Verifies | Explanation of Output | How This Confirms the Requirement |
|---|---|---|---|
| **ping 10.1.1.1** | Checks client (PC) connectivity to the gateway Received | successful ICMP responses on the default gateway | Checks proper DHCP operation, local network connectivity. |

```
PC1> ip dhcp
DORA IP 10.1.1.2/26 GW 10.1.1.1

PC1> show

NAME   IP/MASK              GATEWAY           MAC                LPORT  RHOST:PORT
PC1    10.1.1.2/26          10.1.1.1          00:50:79:66:68:02  10076  127.0.0.1:10077
       fe80::250:79ff:fe66:6802/64

PC1> ping 10.1.1.1
84 bytes from 10.1.1.1 icmp_seq=1 ttl=255 time=15.225 ms
84 bytes from 10.1.1.1 icmp_seq=2 ttl=255 time=16.333 ms
84 bytes from 10.1.1.1 icmp_seq=3 ttl=255 time=15.386 ms
84 bytes from 10.1.1.1 icmp_seq=4 ttl=255 time=17.043 ms
84 bytes from 10.1.1.1 icmp_seq=5 ttl=255 time=15.580 ms

PC1>
```

# SECTION 3: STRATEGY FOR IMPLEMENTATION PLAN AND TROUBLESHOOTING

## 3.1 Implementation Plan

The implementation strategy takes a structured bottom-up approach that has started with basic infrastructure and then progressed to include higher-level services. This sequenced approach helps to address interdependencies of configuration, reduces risk and enables systematic validation at each point which helps faults to be detected early and not to go further into more complex configurations. The process is divided into six logical phases spread out for around five days, each of which is dependent on the successful completion of its predecessor.

The first phase focuses on the setting up of basic infrastructure and IP addressing. During this stage all routers are configured with hostnames, Cisco Express Forwarding (CEF) is enabled to ensure that the packets are forwarded optimally and physical interfaces are provides with IP addresses following the addressing plan. Point-to-point links between routers are configured first in order to protect the backbone connectivity, and then LAN-facing interfaces are configured. Loopback interface is advertised on all provider edge (PE) routers and the distribution router(P) and are used as stable identifiers with routing protocols such as OSPF & BGP protocols Customer edge (CE) routers are populated with some basic static routes towards their respective PE routers. Validation at this point uses interface summary for interface status validation and pings to specific interfaces for adjacency validation. Establishing reliable Layer 3 connectivity in the first place means that if there are any errors in the cabling, addressing, or physical configuration, they will be detected right away as opposed to the issues being masked by the higher layer protocols.

The second phase consists of introducing the MPLS core infrastructure. Upon verifying the IP connectivity, OSPF Area 0 is configured between all provider routers and PE to PE links to provide dynamic reachability information between loopback interfaces. Following OSPF convergence, MPLS is enabled globally as well as all provider facing interfaces. The Label Distribution Protocol (LDP) is configured to use loopback interfaces as router identifiers, which further improves the stability and the resilience of the session. Verification focuses on verifying the OSPF adjacencies, reachability to the PE loopbacks, LDP neighbor establishment and verifying that the labels are correctly assigned within the MPLS forwarding table. Implementing as a separate phase the MPLS core allows to focus on troubleshooting IGP and label distribution problems without the extra complexity of VPN services.

The 3rd phase is Virtual Routing and Forwarding (VRF) instances and multiprotocol BGP (MP-BGP) implementation to support VPN routing. VRF instances are created on PE routers with the help of suitable route distinguishers and route targets. For full mesh connectivity requirements, identical route targets are used while isolated VPN requirements are enforced using different route targets. The customer facing interfaces are linked with respective VRF and MP-BGP are configured between all PE routers using the loopback addresses as the update sources. The address family VPNv4 is enabled with extended communities to be used for route-target exchange. Static customer routes are static to each VRF & redistributed into BGP. Verification includes verifying VRF definitions, verifying BGP Peer Establishment, verifying VPN Route Exchange, and end to end connectivity using VRF specific ping commands. Implementing VPN routing separately ensures that route distribution works properly before the policies of encryption or access control are introduced.

Sensitive communications are then IPsec encrypted on the fourth phase. Encryption is introduced intentionally, and with the understanding that unencrypted connectivity should function correctly, so that it is easy to distinguish between routing problems and encryption problems. Robust IPsec parameters like AES-256, SHA-256, and Diffie-Hellman Group methacrylate in tunnel mode are used since then, when lighter parameters are used than transport mode for less relevant traffic. Cryptographic access control lists (ACLs) are unambiguously defined so that they match only the traffic that should be permitted, leaving non-cryptographically protected traffic alone. Verifying includes checking ISAKMP and IPsec security associations as well as checking packet encryption counters and tunnel-tested control. Implementing the IPsec at this stage isolates the encryption-related failures and makes troubleshooting of them simpler.

The fifth phase brings the access control and security policies. ACLs are set up to enforce traffic restrictions such as reflexive ACLs that perform access based on traffic initiated by a session and standard ACLs that explicitly block unauthorized access. ACLs are applied in directions that bring maximum efficiency and clarity. Verification with the help of ACL match counters, testing permit and deny scenarios for ACL, validation of reflexive session creation. Implementing security controls after the routing and the encryption services are green allows one to ensure that any access failures can be attributed to ACL logic as opposed to underlying connectivity issues.

The final phase allows DHCP to be run and thorough testing. DHCP scopes are set up with suitable exclusions for gateways and servers and the end devices are checked to get addresses dynamically. A complete testing matrix ensures testing of all requirements simultaneously, so as to ensure that no unwanted interactions occur between services. Performance and stability testing ensures the network is functioning as expected operationally. DHCP is used last of all to avoid complications of dynamically changing IP addresses during earlier phases of troubleshooting.

| Phase | Implementation Focus | Key Technologies | Validation Method |
|---|---|---|---|
| **Phase 1** | Base infrastructure & IP addressing | IP addressing, CEF, static routing | show ip interface brief, ping to adjacent routers |
| **Phase 2** | Provider core deployment | OSPF, MPLS, LDP | show ip ospf neighbor, show mpls ldp neighbor, show mpls forwarding-table |
| **Phase 3** | VPN routing | VRF, MP-BGP (VPNv4), RT/RD | show ip route vrf, show ip bgp vpnv4 all, VRF-aware ping |
| **Phase 4** | Secure communication | IPsec (ISAKMP, ESP) | show crypto isakmp sa, show crypto ipsec sa |

## 3.2 Troubleshooting Strategy

The troubleshooting methodology is quite a structured approach to troubleshooting based upon the OSI model to move down the chain methodically from the bottom to the top. This paradigm saves unnecessary effort on high tech diagnostics if the base level connectivity problems exist. During the preliminary phases physical and data link issues are solved through scrutinizing interface status, error counters and hardware connectivity: Issues such as cabling faults or shutdown interfaces are resolved immediately. OSPF adjacency issues will result in areas of mismatch, inconsistencies on MTUs, and matching authentication to diagnose the issue, although some typical reasons are neighbor states, interface configurations, and debug outputs.

The focus of MPLS troubleshooting is checking the LDP neighbor establishment and label distribution. Missing LDP sessions normally means that MPLS is neither enabled over interfaces, that conflicts with a router ID or missing IGP reachability. Ensuring CEF activation and advertisement of loopbacks solves most problems.

For the case of VRF and MP-BGP issues, neighbor states are analyzed in order to identify TCP connectivity failures, configuration errors, or address-family errors. VPN route visibility is checked for in the BGP and VRF routing tables with route-target mismatches and missing extended community exchange common failure points.

Phase 1 and phase 2 failures can be thus distinctly separated when troubleshooting IPsec. Policy mismatches, wrong pre-shared keys, and unreachable peers are common causes of Phase 1 failures and transform-set mismatches and wrong crypto-map application are common causes of Phase 2 problems. Packet counters are used to verify matching of traffic crypto ACLs.

Finally, access control related issues are diagnosed by analyzing ACL counters, interface application direction, and the order of rules. Unauthorized traffic or blocked legitimate traffic is solved by polishing ACL logic and making sure of correct reflexive

ACL pairing. This serial and systematic approach to troubleshooting ensures quick troubleshooting and efficient resolution across the entire implementation lifecycle.

| **Failure Area** | **Symptom Observed** | **Diagnostic Command** | **Interpretation of Output** | **Corrective Action** |
|---|---|---|---|---|
| **Interface / Layer 3** | No connectivity to neighbour | show ip interface brief | Interface down or incorrect IP | Enable interface / correct addressing |
| **OSPF** | No adjacency | show ip ospf neighbor | Stuck in INIT/EXSTART | Fix MTU, authentication, area mismatch |
| **MPLS** | VPN sites unreachable | show mpls ldp neighbor | LDP session missing | Enable MPLS, verify IGP reachability |
| **VRF routing** | Missing VPN routes | show ip route vrf <VRF> | Route not present | Fix VRF binding or static routes |
| **MP-BGP** | No VPN route exchange | show ip bgp vpnv4 all | Routes not imported/exported | Correct RT/RD configuration |
| **IPsec Phase 1** | Tunnel not established | show crypto isakmp sa | No active ISAKMP SA | Fix pre-shared key or peer reachability |
| **IPsec Phase 2** | No encrypted traffic | show crypto ipsec sa | Zero packet counters | Fix crypto ACL or transform-set |
| **ACL enforcement** | Legitimate traffic blocked | show access-lists | Hits on deny rules | Refine ACL order or logic |
| **DHCP** | Client not receiving IP | show ip dhcp pool | Pool exhausted or misconfigured | Adjust pool or exclusions |

References:


Abdullah, A.M. (2017) 'Advanced Encryption Standard (AES) Algorithm to Encrypt and Decrypt Data', Cryptography and Network Security, 16(1),p. 11.

Akintola, G. B., & Falade, P. V. (2024) ‘Evaluating the performances of digital data transmission links on a client-server network model’, *International Journal of Scientific Research in Computer Science and Engineering*, 12(2).

Bilhaj, S., Ahmed, N., & Ashtawi, A. (2026). Design and implementation of a secure WAN using site-to-site VPN: A practical comparison with MPLS. Alqalam Journal of Medical and Applied Sciences, 9(1), 46-51.

Chen, W., 2025. Architecture Design and Performance Analysis of MPLS VPN in Meteorological Wide-Area Networks. International Journal of Advance in Applied Science Research, 4(8), pp.119-124.

Dharun, M.R., AS, A.K., Bharatula, S.D. and Sasikumara, G., 2024, December. Implementing Border Gateway Protocol in MPLS Using Layer 3 VPN. In 2024 International Conference on Smart Electronics and Communication Systems (ISENSE) (pp. 1-5). IEEE.

Frankel, S., Kent, K. and Lewkowski, R. (2005) 'Guide to IPsec VPNs',NIST Special Publication 800-77. Gaithersburg, MD: National Instituteof Standards and Technology.

Kent, S. and Seo, K. (2005) RFC 4301: Security Architecture for theInternet Protocol. Internet Engineering Task Force.

Mungse, M.M. (2023) 'Secure Network Design and Implementation using MPLS VPN', MSc Research Project. Dublin: National College of Ireland.

Mungse, M.M., 2025. Secure Network Design and Implementation using MPLS VPN (Doctoral dissertation, Dublin, National College of Ireland).

Ojo, S.A., Thompson, A.F.B., Iyare, M.O. and Alese, B.K. (2015) 'OnInformation Integrity Measurement with Secure Hash Algorithm (SHA)',In SITE 2015: Informing Science + IT Education Conferences: USA,pp. 327-338.

Okoh, C., Theophilus, W. N., Dawkins, P., & Paheerathan, S. (2024) ‘Enhancing data security through VLSM subnetting and TCP/IP model in an ENT’, *Applied Sciences*, 14(23), p. 10968.

Perez, E.L., Watteyne, T., Marin-Lopez, R., Onete, C., Papon, C. andVučinić, M. (2025) 'Pre-Shared Key Authentication With EDHOC: TheSecurity–Performance Tradeoff', IEEE Access, 13, pp. 200367-200381.

Sofi, U.B. and Gurm, E.R.K., 2015. Comparative Analysis of MPLS Layer 3vpn and MPLS Layer 2 VPN. IJCST, 3(3), pp.214-220.